Nikos Pagonas

Contact Email: nikospagonas00@gmail.com — Website: nikpag.github.io

Education

Columbia University Sep 2024 – Present
PhD in Computer Science New York, NY, US
Advisor: Prof. Kostis Kaffes
GPA: 3.94/4.00

Member of DAPLab, focusing on improving the performance and efficiency of agentic serving. I build systems that exploit workflow structure and runtime characteristics to optimize model routing, scheduling, and resource allocation.

National Technical University of Athens (NTUA) Oct 2018 – Jun 2024
Master of Electrical and Computer Engineering Athens, Greece
Advisors: Prof. Nikos Vasilakis, Prof. Georgios Goumas
GPA: 9.39/10.00 (14^{th} out of 344 students)

Thesis: *"SPLaSh: Scaling Out Shell Scripts on Serverless Platforms"*

Experience

Microsoft Research May 2026 – Aug 2026
Research Intern Redmond, WA, US
Supervisor: Sharad Agarwal

Analyzed production traces to motivate latency awareness in the Microsoft Foundry model router. Worked on adding model-routing support to an LLM serving simulator used by Azure CoreAI, aiming for higher-fidelity opportunity analysis through production-trace replay.

Google May 2025 – Aug 2025
Student Researcher Sunnyvale, CA, US
Supervisor: Prof. Arvind Krishnamurthy

Designed and implemented Cortex, a prototype agentic serving platform that leverages workflow awareness and stage isolation to improve scheduling and KV cache efficiency in agentic NL2SQL workflows.

Brown University Sep 2023 – May 2024
Student Researcher Providence, RI, US
Advisor: Prof. Nikos Vasilakis

Contributed to PaSh, an open-source Linux Foundation project. Worked on improving the performance of shell scripts by running them on serverless infrastructure, and providing fault tolerance for distributed shell script execution.

Arrikto Inc. Dec 2022 – Mar 2023
Software Engineering Intern Athens, Greece

Supervisor: Vangelis Koukis

Explored how to serve PyTorch models more efficiently through KServe, a highly scalable machine learning deployment platform for Kubernetes.

Awards

Columbia University Presidential Fellowship 2024 – Present

A.G. Leventis Foundation Scholarship 2025 – Present

Gerondelis Foundation Scholarship 2025

ACM SOSP 2025 Student Travel Grant 2025

Publications

VineLM: Trie-Based Fine-Grained Control for Agentic Workflows
Nikos Pagonas, Matthew Lou, Tianyi Peng, Dan Rubenstein, Kostis Kaffes

Agentic Data Environments
Elaine Ang, Chenxi Huang, Georgios Liargkovas, Jerry Liu, Jinhui Liu, Nikos Pagonas, Charlie Summers, Haonan Wang, Jiakai Xu, Tianle Zhou, Yusen Zhang, Zhou Yu, Zhuo Zhang, Tianyi Peng, Kostis Kaffes, Eugene Wu
IEEE Data Engineering Bulletin, March 2026

AgentOpt v0.1 Technical Report: Client-Side Optimization for LLM-Based Agents
Wenyue Hua, Sripad Karne, Qian Xie, Armaan Agrawal, Nikos Pagonas, Kostis Kaffes, Tianyi Peng

Cortex: Workflow-Aware Resource Pooling and Scheduling for Agentic Serving
Nikos Pagonas, Yeounoh Chung, Kostis Kaffes, Arvind Krishnamurthy
SOSP 2025 Workshop on Systems for Agentic AI (SAA 2025)

Which Model for Which Role? Sample-Efficient Agent Configuration Search
(Under Submission)

λeash: Scaling Out Shell Scripts with Reliable Cloud Computing
(Under Submission)

Service

Artifact Evaluation Committees
ASPLOS 2027, EuroSys 2027, ATC 2026, CAIS 2026, EuroSys 2026, SOSP 2026, ATC 2025, OSDI 2025, SOSP 2025

Web and Publication Chair
EuroSec 2024

Teaching

Columbia University Fall 2025
Teaching Assistant New York, NY, US
Course: Networks, Crowds, and the Web
Professor: Augustin Chaintreau

Columbia University Spring 2025
Teaching Assistant New York, NY, US
Course: Agentic Systems Made Real
Professors: Eugene Wu, Kostis Kaffes

Brown University Fall 2024
Teaching Assistant Providence, RI, US
Course: Systems Transforming Systems
Professor: Nikos Vasilakis

Guided students through a systems research project, including implementation, evaluation, and paper writing. Paper submitted to USENIX ATC 2025.

Brown University Spring 2024
Teaching Assistant Providence, RI, US
Course: Distributed Systems
Professor: Nikos Vasilakis

Helped revamp the course by redesigning its semester-long project and implementing its autograding infrastructure from scratch.

National Technical University of Athens Spring 2022
Lab Assistant Athens, Greece
Course: Operating Systems
Professors: N. Koziris, G. Goumas

National Technical University of Athens Spring 2020
Lab Assistant Athens, Greece
Course: Programming Techniques
Professors: N. Papaspyrou, A. Pagourtzis, G. Stamou, G. Goumas

National Technical University of Athens Fall 2019
Lab Assistant Athens, Greece
Course: Introduction to Programming
Professors: N. Papaspyrou, D. Fotakis, S. Zachos

Research Mentoring

Armaan Agrawal May 2026 – Present
Columbia University, BS

Grace Li Sep 2025 – Present
Columbia University, BS

Matthew Lou Sep 2025 – Jun 2026
Columbia University, BS → next: Hudson River Trading

Joey Huang Sep 2025 – Dec 2025
Columbia University, BS → next: Google

Invited Talks

VineLM: Trie-Based Fine-Grained Control for Agentic Workflows May 2025
Languages, Systems, and Data Seminar @ University of California, Santa Cruz
Host: Lindsey Kuper

Skills

Prog. Languages	Python, C, Shell, Javascript, Java, Go
Frameworks	LangChain, LangGraph
Platforms	AWS, Docker, Kubernetes, Apache Kafka, Apache Spark
Tools	Git, LaTeX

Languages

English	Fluent — TOEFL iBT 116/120 (C2)
Greek	Native